

Garrett Hagen

Software Engineer

✉ garretthagen21@gmail.com

🔗 github.com/garretthagen21

📞 (412) 897-5699

📍 San Diego, CA

EXPERIENCE

Apple Inc — TV App

iOS Software Engineer · July 2022 – Present

- Delivered Apple TV App functionality for Apple Intelligence-powered Siri through cross-org collaboration with the Siri and Intents frameworks teams.
- Own the TV Home Screen widget, shipping new features and improvements across Apple platforms.
- Built client support for TV App features, including condensed and full sports replays and partner-app integration features.
- Own the client instrumentation that measures app launch and page-load performance and reports UI events across the TV App, refactoring it to adopt standardized internal frameworks.
- Led the UI and data-modeling layer for an upcoming iOS feature.

Dexcom Inc — Continuous Glucose Monitoring Systems

iOS Engineer, New Markets Team · Aug 2021 – July 2022

- Owned transmitter integration (G6 and G7) for the hospital CGM iOS app, maintaining and integrating Dexcom's existing Bluetooth LE communication frameworks.
- Built and improved areas of the patient onboarding flow.

QA Engineer, Transmitter Team · Aug 2020 – Aug 2021

- Wrote automated tests exercising real-world usage scenarios between CGM transmitters and mobile apps over Bluetooth LE.
- Built an Appium suite automating typical usage scenarios to reproduce and analyze BLE connection drops, plus tooling to inspect the captured data.

Gener-8 — FDA-Cleared CPM Self-Rehabilitation Devices

Software Development & Hardware Design · 2016 – Present · Part-time

Second-generation system · in progress

- Built an agentic development harness (Ralph loop) coordinating multiple Claude agents across the BLE peripheral and a paired iOS device. One agent SSHes into the Gener-8 to tail logs and run commands while another drives and monitors the iOS app, enabling automated, remote development and debugging.
- Refactored the Gener-8 website using agentic, AI-assisted engineering workflows.
- Redesigned the machine hardware around a Raspberry Pi system and developed multi-threaded firmware in Python.
- Added real-time calibration offsets, improved LED feedback, and an extensive Bluetooth LE feature set: headless Wi-Fi configuration, wireless firmware updates, motor-speed control, sensor-state reading, rep counting, and task-progress updates.
- Moved the app's client logic behind Firebase Cloud Functions endpoints, centralizing business logic and thinning the client.
- Migrated the codebase from CocoaPods to Swift Package Manager, splitting shared code into reusable Gener-8 packages used by both the factory-setup and production apps.
- Wrote Python admin tools and manufacturing software for the platform, spanning system database administration and Raspberry Pi device provisioning.

First-generation system

- Designed the circuitry and embedded software for an on-device angle-calculation system that streams real-time IMU measurements over Bluetooth LE.
- Shipped the Gener-8 iOS app, covering machine communication, calibration, patient workout-data capture and storage, patient-therapist messaging, and rehab visualization and feedback.
- Owned the full technical stack, from circuit design and manufacturing to app development and App Store deployment.

Carnegie Robotics — Advanced Robotics Sensors & Platforms

Software Co-op · 2018 – 2019

- Created and maintained a sizeable C++ framework that reduced boilerplate for projects using stereo-camera functionality.
- Built a C++ application assessing extrinsic/intrinsic calibration quality for MultiSense cameras using OpenCV, Ceres-Solver, Boost, and GTest.
- Developed an interactive Qt/OpenCV C++ GUI for concurrent grayscale, color, and disparity streaming across all MultiSense camera models.
- Contributed embedded C firmware and Python tooling for MultiSense PCB verification, ROS/OpenCV data-collection utilities, and Docker/CMake build and deployment improvements.

PATENTS

Designed and patented the **Durabat®** baseball training device, used by professional players.

EDUCATION

University of Pittsburgh

BS, Computer Engineering · 2016–2020

- Major GPA 3.82 · Cumulative 3.55
- Engineering Dean's Honor List (2017–2020)

PROJECTS + OPEN SOURCE

Logician 2026 · in development

- Actively developing a voice studio engineer for Logic Pro, built agentially: a native SwiftUI macOS app with its own Claude agent loop and on-device speech recognition, controlling Logic through a forked Logic Pro MCP server.

VizRock 2026

- Raspberry Pi show controller that runs live shows from performers' MIDI footswitch cues, launching Resolume visuals over OSC and controlling peripheral LED lighting through ESP32 nodes over ESP-NOW, all managed from a browser UI. Developed agentially across the Pi app and firmware.

wpa-pyfi 2021

- Open-source Python package that adapts and extends an existing framework to programmatically manage Raspberry Pi Wi-Fi connections and settings from the CLI or as a library. Published on PyPI.

OmniBot 2020

- Omnidirectional robot built with a student-led team for a senior capstone. Developed the iOS app and Bluetooth LE control protocol (joystick, autopilot, and gesture control), training an on-device image-recognition model on hand gestures that drove the robot from the phone's camera feed.

Biometric Lock 2019

- Swift iOS app communicating over Bluetooth LE with an Arduino-driven fingerprint scanner and solenoid lock. Designed all software and circuitry and rendered a custom enclosure mounted in the door frame.

MUSIC + ART

Lead stage production, content and merch for my bands. Run live visuals in Resolume cued by VizRock, design in Canva and CapCut, and track and arrange demos in Logic Pro.

VOLUNTEERING

HCEF Instructor 2019–2020

- Taught basic computer science to middle-school students experiencing homelessness.